MARTIN DESSART

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Master's student specialized in ethology and cognition, with an engineering background. Eager to start a PhD in experimental psychology.

Education

UNIVERSITY OF STRASBOURG
Strasbourg, France
09/2018 – 09/2020

MSc WITH HIGH HONOURS
Ecophysiology – Population genetics
Neurobiology of rhythms – Functional and evolutive ecology
Ethology, cognition and behavior – Statistics analysis using R

UDACITY & GOOGLE
Distance learning
Online Exams
11/2017 – 08/2018

FRONT-END WEB DEVELOPER DEGREE
Formation HTML – CSS – JavaScript - Frameworks
Project design – Website creation – Database
Community support

INSTITUT CATHOLIQUE DES ARTS ET METIERS
Toulouse, France
09/2014 – 09/2017

ENGINEER'S DEGREE
Software and electrical engineering
Project management – Human resources
Finance – Industrial strategy

INSTITUT CATHOLIQUE DES ARTS ET METIERS
Toulouse, France
09/2012 – 09/2014

PREPARATORY CLASS
Electrical engineering
Web technologies – Networks management
Industrial organization

Research Experience

CRCA, GEODE
CNRS University Paul Sabatier
Toulouse, France
07/2020 – 10/2020

VOLUNTEER INTERNSHIP
Interaction between pollinators and invasive floral species
Field: use of spectrophotometer to collect reflectance data
Data treatment using honeybee and bumblebee vision models

ANIMAL MINDS LAB
The University of Auckland
Christchurch, New Zealand
01/2020 – 07/2020

MASTER'S FINAL PROJECT: Evolutive cognition in kea
Design of a cognitive experiment using Judgment Bias Test
Bioacoustics data record and treatment using Raven and Praat
Bayesian statistics models using MCMC method

UMR MARBEC
Pole CNRS
Montpellier, France
04/2019 – 06/2019

RESEARCH PROTOCOL DESIGN | PhD Candidate assistant
Design and field experiment on gut microbiota sampling
Methane measurement by Gas Chromatography
DNA extraction and PCR on methanogens

ASSOCIATION MELINDIKA
Kafue National Park, Zambia
03/2018 – 04/2018

VOLUNTEERING – VET ASSISTANT
One month: Treating cattle and interview farming communities
3 weeks Project: Fencing in Zambia
Designed a new fencing system to protect cattle from predators

AUCKLAND UNIVERSITY OF TECHNOLOGY
Auckland, New Zealand
09/2016 – 02/2017

ENGINEERING THESIS | PhD Candidate assistant
Study of electric wind farms in New Zealand
Longitudinal analysis using Matlab
Organization of industrial meetings

Engineer Experience

CLEMESSY AERONAUTICS & SPACE
Toulouse, France
09/2014 – 09/2017

APPRENTICE ENGINEER
Project: Physical and virtual integration of automatic cable tester
Customer/supplier relationship management
Design, planning and supervision of electrical tests

TESEO
CLEMESSY Entity
Torino, Italy
05/2015 – 08/2015

INTERN IN COMPUTING DESIGN
Process design using Excel VBA Macro
Electrical cost design for customers
Radars and electronic cards calibration in anechoic chamber

Other Skills and Experience

LANGUAGES
English: fluent - TOEIC 840 – 2015
Spanish: read and speak with basic competence

COMPUTING
Website design: HTLM5 - CSS3 – JavaScript
Programming: VBA – Matlab – Python - SQL

VOLUNTEERING
*09/19-Present: **GEPMA** – Study and protection group for mammals in Alsace*
Field work and formation (*meles meles, felis sylvestris sylvestris)*

VOLUNTEERING
01/18-02/18: **AVA Sanctuary, France**
Animal keeper assistant and PhD Candidate assistant: cat biological rhythms
Telemetry, GPS and RFID set-up

VOLUNTEERING
2014 – 2017: **Association Terres en Mêlées**
Activity leader in France, Morocco and Madagascar
Marketing and outreach
Creation of advertizing events

VOLUNTEERING
10/16-12/16: **Auckland Zoo**
Team « Enrichment Behavior »
Design of reflection games for primates

Scolarships

2017 UDACITY Scholarship – 2200 USD
2019 Travelling grant: Region Alsace GrandEst – 2640 euros

Conference presentations

Aug 2020 Research Center on Animal Cognition - University Paul Sabatier -Toulouse III
"Effects of the Warble Call on Optimism in Kea (*Nestor notabilis)"*

Master's thesis abstract

TITLE: A measurement of optimism in kea (nestor notabilis)

KEYWORDS: judgment bias, emotional state, nestor notabilis, animal welfare, parrot, vocalisations, emotion

ABSTRACT :

How important is the link between vocalisations and internal emotional states? Emotional state influences cognition by three types of cognitive biases: attention, memory and judgment biases. The judgment bias test (JBT) has been introduced to measure animals' own judgment as an indicator of their emotional state and welfare.

Here, we created a paradigm using a visual JBT where we tested the effect of two vocalisations (warble call as a playful call, hence "positive-valenced" and trill call as an alarm call, hence "negative valenced") on 8 kea (N*estor notabilis*). For this experiment, kea were trained to distinguish the presentation of a positive (rewarding) or a negative (non-rewarding) cue, and then tested on novel ambiguous to probe their judgment of them. Simultaneously with the introduction of each ambiguous cue, kea heard a playback of a vocalisation.

We measured the latency to approach the cue for one positive, one negative and three types of ambiguous cues, using colored flap doors (0%, 20%, 40%, 60% and 80% grey) and a specific apparatus. Using Bayesian multilevel analysis, we did not find strong differences in latency between the two vocalisations conditions. Yet, kea showed a remarkable learning capacity and good consistency throughout the experiment.

Despites the unpredictable behavioural inter-individual differences of kea, this study provides robust support to set up JBT test on Psittacidae and calls for the analysis of animal welfare through the lens of animal emotional state as a support of other most commonly used tests.

References

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